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Transient Flow to a Large-Diameter Well in an Aquifer With Storative Semiconfining Layers

ALLEN F. MOENCH

U.S. Geological Survey, Menlo Park, California

The Hantush theory of leaky aquifers with storge in the semiconfining layers is combined with large-diameter well theory to produce equations that can be used in the analysis of pumped-well and observation well data for stratified formations. Included in the equations are storage in the pumped well and a linear resistance to flow at the sand face or well bore skin. Three cases proposed by Hantush are considered. These depend upon whether the upper boundary of the overlying semiconfining layer or the lower boundary of the underlying semiconfining layer are constant head or no-flow boundaries. Laplace transform solutions, valid for the complete time domain, are given for each of the three cases for the hydraulic head in the pumped well, the aquifer, and each of the semiconfining layers. Type cures obtained by numerical inversion are selected to illustrate the effects of well bore storage, well bore skin, and leakage. Although several dimensionless parameters are involved, these parameters tend to influence the character of different portions of the type curves, suggesting that unique matches are possible. The type curves show that well bore storage in a large-diameter well may completely obliterate effects of leakage derived from compressible storage in semiconfining layers. For the purposes of aquifer testing, it may be possible to reduce the magnitude of well bore storage in a large-diameter well and thus reveal the presence of leaky semiconfining layers. This may help to prevent erroneous interpretation of the well test data and incorrect evaluation of the aquifer parameters.

INTRODUCTION

Accurate prediction of aquifer response to pumpage from wells in sedimentary formations requires a knowledge of the hydraulic properties of not only the producing aquifer but also the associated overlying and underlying semiconfining layers. It is generally recognized that low-permeability layers, or aquitards, adjacent to the main pumped aquifer may be composed of highly porous and compressible clay materials and may therefore be capable of contributing substantial quantities of stored water to the aquifer. Improved estimates of the hydraulic properties of the aquifer and adjacent semiconfining layers can be obtained with the help of improved analytical and numerical techniques.

Present-day methods of well test analysis began with *Theis* [1935]. The well-known Theis equation for the lowering of the piezometric surface due to constant well discharge applies to a homogeneous and isotropic aquifer of infinite areal extent, confined above and below by impermeable layers. Because of deviations noted in field tests from the value predicted by the Theis equation, it became apparent that confining layers and distributed discontinuous layers contribute water to the pumped aquifer. To account for the contribution of water from confining layers, *Hantush and Jacob* [1955] presented a theory in which it was assumed that leakage through an overlying (or underlying) layer adjacent to the aquifer was proportional to the difference in hydraulic head between the aquifer and a constant-head source bed. This theory does not account for water released from storage in the compressible semiconfining layers and, as a consequence, predicts responses that tend to overestimate the drawdown at early time. Although more advanced theory that includes storage in the semiconfining layers has been available for many years, it appears that the Hantush and Jacob theory is still commonly used, perhaps because of its simplicity and because of the availability of type curves. Unfortunately, its use can easily lead to erroneous evaluation of aquifer and semiconfining layer hydraulic properties.

Omission of storage in the confining layers in *Hantush and Jacob*'s [1955] theory was corrected by *Hantush* [1960]. As a result the early-time drawdown was found to be significantly delayed. *Hantush* [1960] proposed three idealized cases. One of these cases was a "three-aquifer" system in which hydraulic heads in the unpumped aquifers, located above and below the pumped aquifer and separated from it by semiconfining layers, remain constant. Early-time and late-time asymptotic solutions were presented. The early-time theory is of great importance to the proper analysis of well test data. It is unfortunate that even today it is frequently ignored. F. S. Riley and E. J. McClelland (U.S. Geological Survey, unpublished manuscript, 1971) developed type curves and outlined procedures for use of the method. The late-time theory is of more limited usefulness because of the assumption of constant head in the unpumped aquifers. To account for drawdown in the unpumped aquifer of a two-aquifer system, *Neuman and Witherspoon* [1969*a*, 1971] presented a still more general theory; however, because of the number of variables involved and the difficulty in producing type curves, the theory is rarely applied. By using the technique described by *Moench and Ogata* [1984] the theory for the two-aquifer system was successfully applied in estimating the vertical hydraulic conductivity of a semiconfining layer in a sedimentary aquifer system [see *Hamlin*, 1982].

Another of the cases proposed by Hantush assumes the overlying semiconfining layer to be bounded above by impermeable material and the underlying semiconfining layer to be bounded below by impermeable material. This case has considerable practical value as it applies not only to stratified formations [*Streltsova*, 1982, 1984] but also to fractured-rock aquifers [*Boulton and Streltsova*, 1977; *Serra et al.*, 1983].

The theories described in the above papers assumed that the pumped well is a line source. This assumption may not be valid if effects of well bore storage in the pumped well are significant. Effects of well bore storage may be especially im-

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portant when the aquifer transmissivity and storage coefficient are small or when the pumped-well diameter is large. *Papadopulos and Cooper* [1967] and subsequently *Papadopulos* [1967] developed a theory that accounts for effects of well bore storage in the pumped well. They presented analytical solutions and type curves for drawdown in and around a large diameter well in a nonleaky artesian aquifer.

Effects of a resistance to flow at the interface between the aquifer and well bore due to casing perforations or formation plugging can sometimes be ascribed to a thin "skin" of low-permeability material adjacent to the well bore. These effects may strongly influence the hydraulic head response in the pumped well and, to a lesser degree, the response in the nearby observation wells. *Agarwal et al.* [1970] and *Sandal et al.* [1978] incorporated well bore skin into the large-diameter well theory for nonleaky aquifers. For the pumped well the theory is identical to the analogous heat flow theory [*Blackwell*, 1954].

Many refinements have been made in the development of the theory of stratified reservoirs to include assorted boundary conditions, effects of multiple aquifers, anisotropy, and partial penetration. Analyses that combine leaky aquifer theory with the effects of well bore storage have been made by *Lai and Su* [1974] and *Abdul Khader and Ramadurgaiah* [1977]. These papers did not consider the effects of storage in the semiconfining layers and hence may not properly describe the early-time response. Using a finite-element model, *Rama Rao* [1980], however, did include storage in the semiconfining layers. He also included effects of partial penetration and non-Darcian flow in the close vicinity of the pumped well. *Weeks* [1977] gave a state-of-the-art review of the theory of well test analysis and made the point that although many analytical solutions exist, not all are in a form that can be readily evaluated and used for data analysis. A remedy to this situation was suggested by *Moench and Ogata* [1981, 1984], who showed that solutions to many complex problems can be readily evaluated by numerical inversion of Laplace transform solutions.

In this paper, mathematical models are presented that combine the leaky aquifer theory of *Hantush* [1960] with the theory of flow to a large-diameter well of *Papadopulos and Cooper* [1967] and *Papadopulos* [1967]. The paper expands upon a previously published report [*Moench*, 1982] by considering the three boundary value problems proposed by *Hantush* [1960]. Solutions for dimensionless drawdown in the pumped well, the aquifer, and the semiconfining layers are given in the Laplace plane and are evaluated by numerical inversion.

CONCEPTUAL MODEL

Figure 1 illustrates the system under study. Semiconfining layers are located above and below the main pumped aquifer. The aquifer and semiconfining layers are assumed homogeneous, isotropic, of constant thickness, and of infinite radial extent. *Hantush* [1960] proposed three models that will be investigated in this study. These depend upon whether the upper boundary of the semiconfining layer that overlies the aquifer and the lower boundary of the semiconfining layer that underlies the aquifer are constant-head boundaries (case 1), no-flow boundaries (case 2), or a combination consisting of a constant-head boundary above and a no-flow boundary below (case 3). Prior to the onset of pumpage the potentiometric surfaces corresponding to each layer are assumed to be horizontal. It is also assumed that flow in the semiconfining layers is vertical and flow in the aquifer is horizontal. For this to be valid the hydraulic conductivity of the aquifer must be far greater than the hydraulic conductivity of the semiconfining layers. *Neuman and Witherspoon* [1969*b*] have examined the assumption in detail using a finite-element model and provide conditions under which the assumption fails to be valid. They concluded that if the hydraulic conductivity of the aquifer is more than 100 times the hydraulic conductivity of the semiconfining layer then the error introduced by this assumption is less than 5% in the semiconfining layer and negligible in the aquifer. For lower hydraulic conductivity contrasts the error introduced by this assumption is correspondingly greater. Also, the error becomes progressively greater with time and with decreased radial distance. The parameters evaluated from pumped-well data alone may be subject to error due to nonvertical components of flow in the aquitard near the pumped well.

It is assumed that a pumped well, screened only within the aquifer, fully penetrates the aquifer and with the onset of pumpage discharges water to the surface at a constant rate. As in the theory of *Papadopulos and Cooper* [1967] the pumped well contains stored water that is accounted for by a well bore storage factor. Well bore skin, commonly used in petroleum engineering [see *Earlougher*, 1977], is also included in the analysis in order to account for well losses due to a zone of altered hydraulic conductivity in the vicinity of the well bore. Such altered material may be due to factors such as the invasion of pores by drilling mud, the buildup of scale, or well stimulation endeavors. The skin is assumed to be infinitesimally thin and therefore to possess no storage capacity. Both the well bore storage and well bore skin factors are assumed to remain constant in a given field test.

The above requirements are in addition to the usual assumptions, discussed by *Freeze and Cherry* [1979], that go into the linearized differential equation for flow in compressible formations, namely, that Darcy's law applies, that fluid and rock properties are constant, and that fluid density gradients are negligible.

ANALYSIS

The conceptual models can be described mathematically by the following boundary value problems.

For the aquifer the governing equation is the radial diffusion equation with source terms:

$$\frac{\partial^2 h}{\partial r^2} + \frac{1}{r}\frac{\partial h}{\partial r} = \frac{S_s}{K}\frac{\partial h}{\partial t} + w' - w'' \qquad r > r_w \tag{1}$$

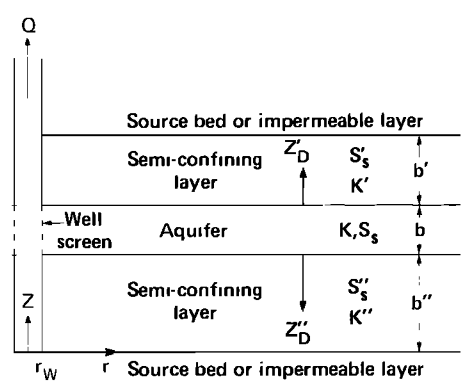

Fig. 1. Schematic diagram of leaky aquifer system.

where

$$w' = -\frac{K'}{Kb}\left(\frac{\partial h'(r,t)}{\partial z}\right)_{z=b''+b}$$

$$w'' = -\frac{K''}{Kb}\left(\frac{\partial h''(r,t)}{\partial z}\right)_{z=b''}$$

The single-primed quantities represent parameters and variables in the overlying semiconfining layer, and the double-primed quantities refer to the underlying layer; w' and w'' represent source terms due to leakage induced by pumping from the aquifer.

The initial condition is

$$h = h_i \qquad r > r_w \tag{2}$$

The boundary conditions are

$$h = h_i \qquad r = \infty \tag{3}$$

$$2\pi r_w Kb\left(\frac{\partial h}{\partial r}\right)_{r=r_w} = Q + C\frac{\partial h_w}{\partial t} \qquad r = r_w \tag{4}$$

where

$$h_w = h - r_w S_w\left(\frac{\partial h}{\partial r}\right)_{r=r_w}$$

Symbols are defined in the notation section.

Boundary condition (4) is the mass balance in the well bore where C is the well bore storage. *Ramey and Agarwal* [1972] pointed out that the effects of well bore storage can be by virture of changing liquid level in the well casing or by virtue of compressibility. Effects of well bore storage are greatest when due to changing liquid level. In this case, $C = \pi r_c^2$, where r_c is the radius of the well casing in the region of the changing liquid level. If the effects are due to compressibility of the liquid and of the well itself in a pressurized well [see *Neuzil*, 1982; *Bredehoeft and Papadopulos*, 1980], $C = V_w \rho_w g C_{obs}$ where V_w is the volume of liquid in the pressurized section, ρ_w is the liquid density, g is the acceleration of gravity, and C_{obs} is the observed compressibility of the combined fluid-well system. S_w is the well bore skin factor. It is a constant of proportionality that relates the flux of water through the skin to the head difference across the skin.

For the overlying semiconfining layer the governing equation is the planar diffusion equation:

$$\frac{\partial^2 h'}{\partial z^2} = \frac{S_s'}{K'}\frac{\partial h'}{\partial t} \qquad b'' + b \le z \le b'' + b + b' \tag{5}$$

The initial condition is

$$h' = h_i' \qquad b'' + b \le z \le b'' + b + b' \tag{6}$$

The lower boundary condition is

$$h' = h \qquad z = b'' + b \tag{7}$$

The upper boundary condition is either

$$h' = h_i' \qquad z = b'' + b + b' \tag{8}$$

or

$$\frac{\partial h'}{\partial z} = 0 \qquad z = b'' + b + b' \tag{9}$$

For the underlying semiconfining layer the governing equation is

$$\frac{\partial^2 h''}{\partial z^2} = \frac{S_s''}{K''}\frac{\partial h''}{\partial t} \qquad 0 \le z \le b'' \tag{10}$$

The initial condition is

$$h'' = h_i'' \qquad 0 \le z \le b'' \tag{11}$$

The upper boundary condition is

$$h'' = h \qquad z = b' \tag{12}$$

The lower boundary condition is either

$$h'' = h_i'' \qquad z = 0 \tag{13}$$

or

$$\frac{\partial h''}{\partial z} = 0 \qquad z = 0 \tag{14}$$

The above equations can be put in nondimensional form by defining the following dimensionless groupings and substituting them into (1)–(14). The resulting equations are given in the appendix.

$$h_D = \frac{4\pi Kb}{Q}(h_i - h) \tag{15}$$

$$h_{wD} = \frac{4\pi Kb}{Q}(h_i - h_w) \tag{16}$$

$$h_D' = \frac{4\pi Kb}{Q}(h_i' - h') \tag{17}$$

$$h_D'' = \frac{4\pi Kb}{Q}(h_i'' - h'') \tag{18}$$

$$t_D = \frac{Kt}{S_s r_w^2} \tag{19}$$

$$r_D = r/r_w \tag{20}$$

$$z_D' = (z - b'' - b)/b' \tag{21}$$

$$z_D'' = (b'' - z)/b'' \tag{22}$$

$$W_D = C/2\pi r_w^2 S_s b \tag{23}$$

Laplace transform solutions to (1)–(14) are derived in the appendix. The solution for dimensionless head in the production well is

$$\bar{h}_{wD} = \frac{2[K_0(x) + xS_wK_1(x)]}{p\{pW_D[K_0(x) + xS_wK_1(x)] + xK_1(x)\}} \tag{24}$$

where p is the Laplace transform variable and

$$x = (p + \bar{q})^{1/2}$$

for dimensionless head in the aquifer the solution is

$$\bar{h}_D = \frac{2K_0(r_D x)}{p\{pW_D[K_0(x) + xS_wK_1(x)] + xK_1(x)\}} \tag{25}$$

Note that the overbar refers to Laplace space. The parameter $\bar{q}$ is a term that represents flow from the semiconfining layers to the aquifer and is determined by solving (5) and (10). In this study the form it takes depends upon the particular combination of boundary conditions (8), (9), (13), and (14). The dimensionless head at points within the semiconfining layer also depends upon the boundary conditions (8), (9), (13), and (14).

The Laplace transform expressions for $\bar{q}$ and the Laplace transform solutions for dimensionless head in the semi-

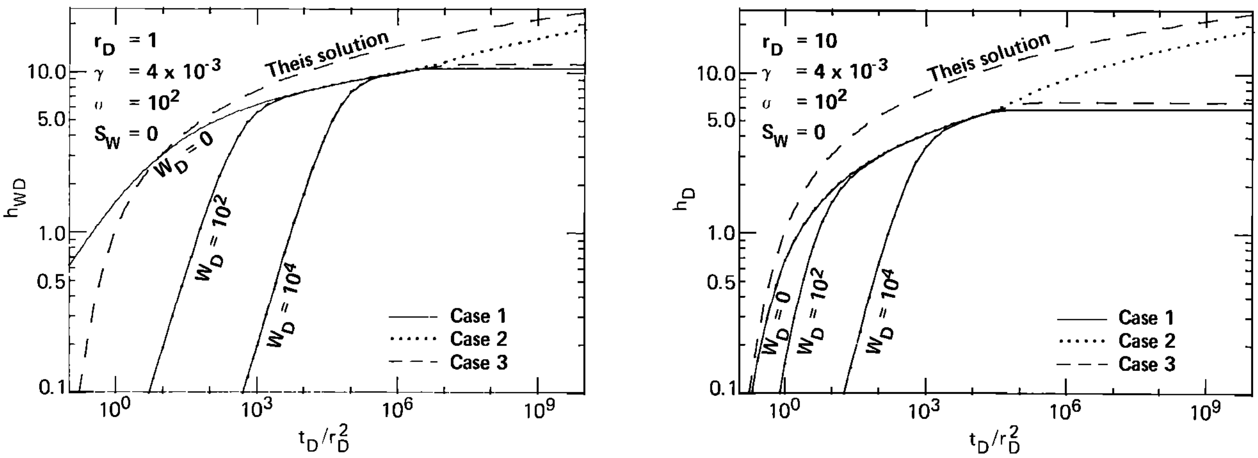

Fig. 2a

Fig. 2b

Fig. 2. Dimensionless drawdown versus dimensionless time for cases 1, 2, and 3 and for the indicated values of γ, σ, W_D, and S_w: (*a*) in the pumped well, (*b*) in the aquifer at $r_D = 10$.

confining layers are written below for each of the three cases proposed by *Hantush* [1960]. Similar expressions using different notation can be found in the work by *Hantush* [1964].

Case 1: boundary conditions (8) and (13)

$$\bar{q} = (\gamma')^2 m' \coth(m') + (\gamma'')^2 m'' \coth(m'') \tag{26a}$$

$$\bar{h}_D' = \bar{h}_D \sinh[(1 - z_D')m']/\sinh(m') \tag{26b}$$

$$\bar{h}_D'' = \bar{h}_D \sinh[(1 - z_D'')m'']/\sinh(m'') \tag{26c}$$

Case 2: boundary conditions (9) and (14)

$$\bar{q} = (\gamma')^2 m' \tanh(m') + (\gamma'')^2 m'' \tanh(m'') \tag{27a}$$

$$\bar{h}_D' = \bar{h}_D \cosh[(1 - z_D')m']/\cosh(m') \tag{27b}$$

$$\bar{h}_D'' = \bar{h}_D \cosh[(1 - z_D'')m'']/\cosh(m'') \tag{27c}$$

Case 3: boundary conditions (8) and (14)

$$\bar{q} = (\gamma')^2 m' \coth(m') + (\gamma'')^2 m'' \tanh(m'') \tag{28a}$$

$$\bar{h}_D' = \bar{h}_D \sinh[(1 - z_D')m']/\sinh(m') \tag{28b}$$

$$\bar{h}_D'' = \bar{h}_D \cosh[(1 - z_D'')m'']/\cosh(m'') \tag{28c}$$

where

$$m' = \frac{(\sigma' p)^{1/2}}{\gamma'} \qquad m'' = \frac{(\sigma'' p)^{1/2}}{\gamma''}$$

$$\gamma' = r_w\left(\frac{K'}{Kbb'}\right)^{1/2} \qquad \gamma'' = r_w\left(\frac{K''}{Kbb''}\right)^{1/2}$$

$$\sigma' = \frac{S_s'b'}{S_s b} \qquad \sigma'' = \frac{S_s''b''}{S_s b}$$

The notation presented above departs from conventional notation. It is used here for consistency with that used by *Moench* [1984] and because it has advantages over conventional notation in that the storage parameters are separate from the hydraulic conductivity parameters and are independent of radial distance. Thus σ and γ are properties of the aquifer/aquitard system. They can be related to the commonly used Hantush parameters, β and r/B, as follows:

$$\beta = \frac{\gamma}{4}(\sigma)^{1/2} r_D \qquad r/B = \gamma r_D$$

The combination of (25) and (26*a*) is identical to the line source Laplace transform solution of *Hantush* [1960, equation (40)] when the term in braces that appears in the denominator of (25) is omitted. Differences in form are due to the dimensionless groupings used in this paper. In the absence of well bore skin and leakage, (24) reduces to the large-diameter well equation of *Papadopulos and Cooper* [1967, equation (10)]. Also, in the absence of well bore storage the combinations of (25) and (26*a*) or (27*a*) are identical to the Laplace transform solutions given by *Hantush* [1964, equations 42 and 43] for a finite-diameter well in a leaky aquifer. In the absence of leakage, but retaining well bore storage and skin factors, (24) is identical to the equation of *Agarwal et al.* [1970], and (25) is identical to the equation of *Sandal et al.* [1978].

The expressions (26*b*), (26*c*), (27*b*), (27*c*), (28*b*), and (28*c*) yield, upon inversion, the theoretical hydraulic head at any point within the semiconfining layers. The average hydraulic head over a finite screened interval within the semiconfining layers is obtained simply by integrating these equations over the desired vertical interval and dividing by the length of the interval. For example, (27*b*) becomes

$$(\bar{h}_D')_{\text{ave}} = \frac{\bar{h}_D\{\sinh[(1 - z_{D2}')m'] - \sinh[(1 - z_{D1}')m']\}}{(z_{D1}' - z_{D2}')m' \cosh(m')}$$

where z_{D2}' and z_{D1}' are the top and bottom of the screen, respectively.

NUMERICAL INVERSION AND VERIFICATION OF THE SOLUTIONS

Analytical inversion of (24) and (25) in their complete forms with nonzero $\bar{q}$ is not easily accomplished, if it is possible at all, for any of the three cases. Even when reduced to the line source solution, analytical inversion yields semi-infinite integral expressions that are difficult to evaluate accurately over the entire time domain of interest (see discussion in the work by *Moench and Ogata* [1984] regarding the results of *Boulton and Streltsova* [1977]). For this reason, *Hantush* [1960, 1964] resorted to asymptotic solutions valid for early or late time. Solutions to (24) or (25) valid for the complete real-time domain can, however, be obtained by numerical inversion. In this paper the *Stehfest* [1970] algorithm is used. The method has been widely used in the petroleum industry for a number

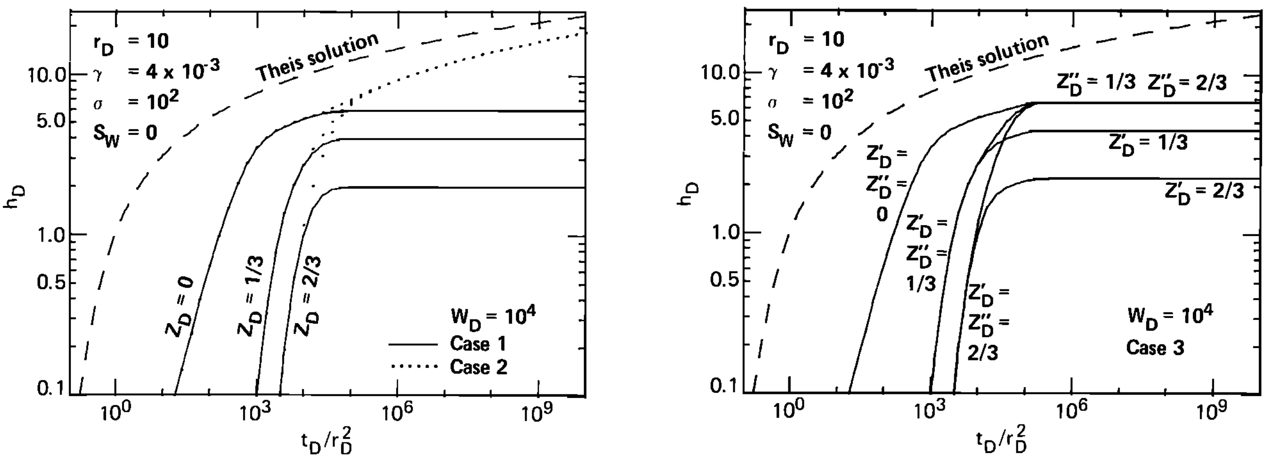

Fig 3. Dimensionless drawdown versus dimensionless time in the aquifer and in the semiconfining layers at $r_D = 10$ for the indicated values of z_D, γ, σ, W_D, and S_w: (a) for cases 1 and 2, (b) for case 3.

of years and has only recently been applied by groundwater hydrologists. *Moench and Ogata* [1984] provided verification of the technique by comparison with tabulations of the *Theis* [1935] solution, the *Hantush* [1960] short-time solution, and the *Papadopulos* [1967] and the *Papadopulos and Cooper* [1967] large-diameter well solution. Results were found to agree to as many significant figures as given in the tables. Results were also found to agree with the finite-difference solution to the problem of an aquifer overlain by a water table aquitard of *Cooley and Case* [1973] and with the type curves given by *Neuman and Witherspoon* [1969*a*] for the two-aquifer problem.

TYPE CURVES

Effects of well bore storage and skin and the influence of leakage from semiconfining layers for cases 1, 2, and 3 are shown graphically in the figures that follow. Dimensionless drawdown in the production well and at various points in the aquifer and semiconfining layers is plotted versus the dimensionless time group t_D/r_D^2 for different values of S_w, W_D, z_D, γ, and σ. In each instance it is assumed that both an upper semiconfining layer and a lower semiconfining layer exist and, for the sake of simplicity in illustrating the results, that they have identical hydraulic properties. For this reason, symbols for the hydraulic properties of the semiconfining layers are left unprimed. Primed quantities appear only where it is necessary to distinguish the upper and lower layers. The line source solution of *Theis* [1935] is plotted as a dashed curve for reference purposes in each figure.

Figures 2*a* and 2*b* show dimensionless drawdown in the production well ($r_D = 1$) and in an observation well at $r_D = 10$ for all three cases assuming negligible well bore skin ($S_w = 0$). The dimensionless well bore storage parameter is assigned the values $W_D = 0$, 10^2, and 10^4. The dimensionless hydraulic flow parameters are assigned the values $\gamma = 0.004$ and $\sigma = 10^2$ (or $\beta = 0.01r_D$).

Differences among the three cases can be seen only at late time when t_D is greater than about 10^6. During "early" time ($t_D < 10^6$) the upper and lower boundaries that distinguish the three cases play no significant role in the solutions, reflecting the fact that leakage into the aquifer is derived entirely from compressible storage in the semiconfining layers. In cases 1 and 3 the drawdown eventually approaches steady state as one might infer from the constant-head boundary condition. The drawdown at late time in case 3 is only slightly greater than in case 1. Case 2 shows that the drawdown increases indefinitely with time. It ultimately parallels the Theis solution and, as it will be shown, is displaced from it to the right by a factor of $1 + 2\sigma$.

The type curves for $W_D = 0$ correspond to the *Hantush* [1964, equations 42 and 43] solutions for cases 1 and 2 in which the well diameter is finite but for which well bore storage is neglected. Type curves for $W_D = 0$ are primarily of academic interest and of little practical value. It can be seen in Figure 2*a* for $W_D = 10^2$ and 10^4 that as t_D becomes small the slope of the type curves approaches unity. This fact was pointed out by *Papadopulos and Cooper* [1967]. It is true for the production well but not for the observation wells.

Figures 2*a* and 2*b* show that large well bore storage may completely mask effects of leakage if the leakage is derived from storage in the semiconfining layers. This is especially true if the parameter σ is small (see Figure 6*a*) or if γ is large (see Figure 6*b*).

It should be kept in mind that the dimensionless well bore storage factor is inversely related to the value of the aquifer storage coefficient and hence may be quite large even when the production well diameter is small. When the well diameter is small, however, because of the definition of dimensionless time ($t_D = Kt/(S_s r_w^2)$) the earliest measurements of drawdown in the production well or nearby observation wells may actually result in large values of t_D, thus placing the data well out on the type curves. Consequently, effects of well bore storage and perhaps early-time leakage derived from storage in the semiconfining layers may not be apparent. Of course, this will also be true for a large-diameter well if aquifer diffusivity K/S_s is large. Thus observable effects of well bore storage are maximized, for a given well diameter, when aquifer transmissivity Kb and the storage coefficient $S_s b$ are small.

Figures 3*a* and 3*b* show the responses that are expected to occur in piezometers located at $r_D = 10$ with $z_D = 1/3$ and $2/3$ for each of the three cases assuming the well bore storage factor $W_D = 10^4$. The remaining parameters have the same values that were used in Figure 2. The aquifer response at $r_D = 10$ ($z_D = 0$) given in Figure 2*b* is reproduced for purposes of comparison.

Figure 3*a* compares cases 1 and 2. The head responses in

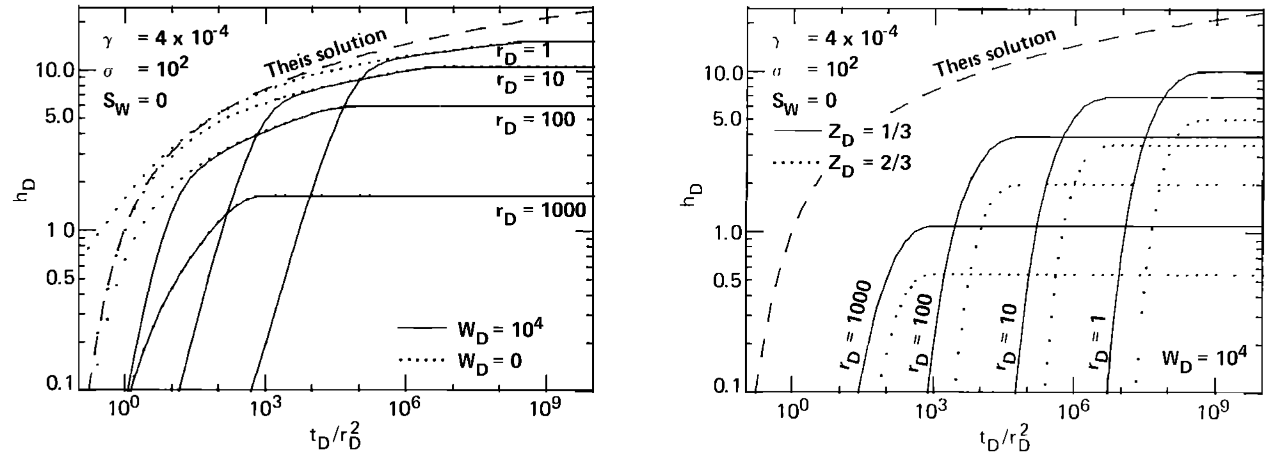

Fig. 4a Fig. 4b

Fig. 4. Dimensionless drawdown versus dimensionless time for case 1 and for the indicated values of r_D, γ, σ, W_D, and S_w: (*a*) for the aquifer, (*b*) for the semiconfining layers at $z_D = 1/3$ and 2/3.

the piezometers in case 1 approach different steady state values depending upon the location of the piezometer within the semiconfining layer. The head responses in the piezometers in case 2 approach the same value as the head in the aquifer. Under this late-time condition, flow from the semiconfining layers no longer derives from compressible storage but from semisteady state flow as described by *Dake* [1978, p. 134].

Figure 3*b* illustrates the more complicated case 3. The head responses in the piezometers located in the lower semiconfining layer, bounded below by an impermeable formation, approach the same value as the head in the aquifer ($z_D' = z_D'' = 0$), and the head responses in the upper semiconfining layer, bounded above by a constant-head source bed, approach different steady state values. The three cases shown in Figure 3 appear identical when t_D is less than 10^6 ($t_D/r_D^2 < 10^4$). Early-time drawdown in either of the two piezometers is not significantly influenced by well bore storage; however, as z_D approaches zero, the piezometer response will approach the aquifer response and will indeed be influenced by well bore storage. It should be recalled that piezometers lcoated in the semiconfining layers near the pumped well are subject to error due to diagonal flow components as described by *Neuman and Witherspoon* [1969*a*]. These errors become more serious with decreased radial distance from the pumped well, increased time, and decreased hydraulic conductivity contrast between the aquifer and the semiconfining layer.

Figures 4*a* and 4*b* show theoretical responses in the aquifer and in the semiconfining layers, respectively, for case 1. Type curves are drawn showing responses as $r_D = 1$, 10, 100, and 1000. The dimensionless hydraulic flow parameters are assigned the values $\gamma = 0.0004$ and $\sigma = 10^2$ (or $\beta = 0.001 r_D$), and it is assumed that the well bore skin is negligible. The type curves in Figure 4*a* for $W_D = 10^4$ cross over one another at early time. That this is due to well bore storage can be seen by comparison with the type curves for $W_D = 0$. At $r_D = 1000$, effects of well bore storage occur at such small values of t_D/r_D^2 that they do not appear on the figure. Figure 4*b* shows piezometer responses in the semiconfining layers at two vertical positions, $z_D = 1/3$ and 2/3, between the aquifer and the source beds. At these positions the responses are sufficiently delayed that effects of well bore storage are not apparent. It is worth noting that as z_D approaches zero, the type curves in Figure 4*b* will approach those in Figure 4*a*.

Figures 5*a* and 5*b* show the theoretical responses in the

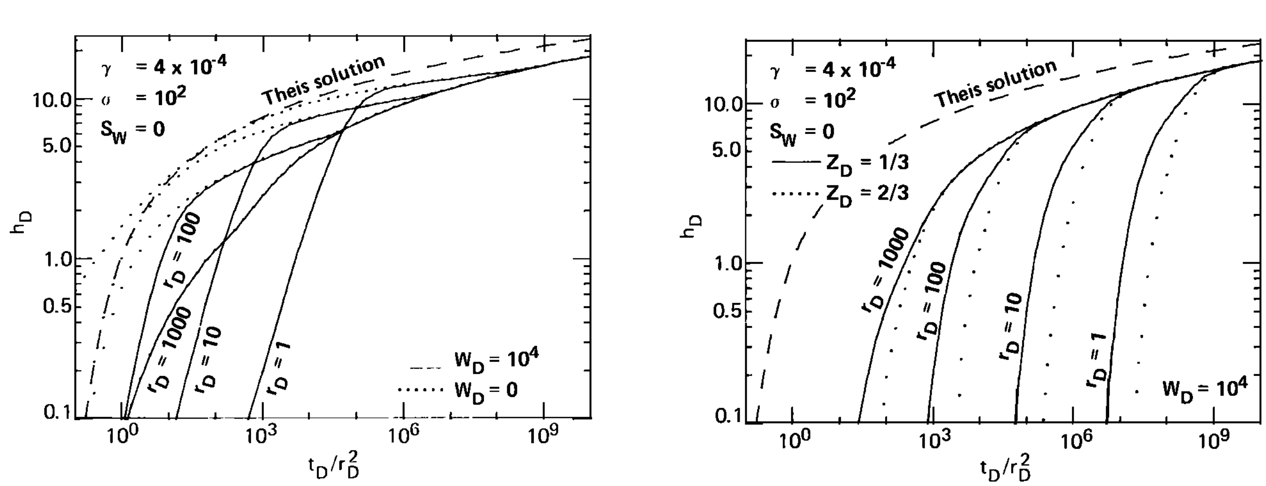

Fig. 5a Fig. 5b

Fig. 5. Dimensionless drawdown versus dimensionless time for case 2 and the indicated values of r_D, γ, σ, W_D, and S_w; (*a*) for the aquifer, (*b*) for the semiconfiing layers at $z_D = 1/3$ and 2/3.

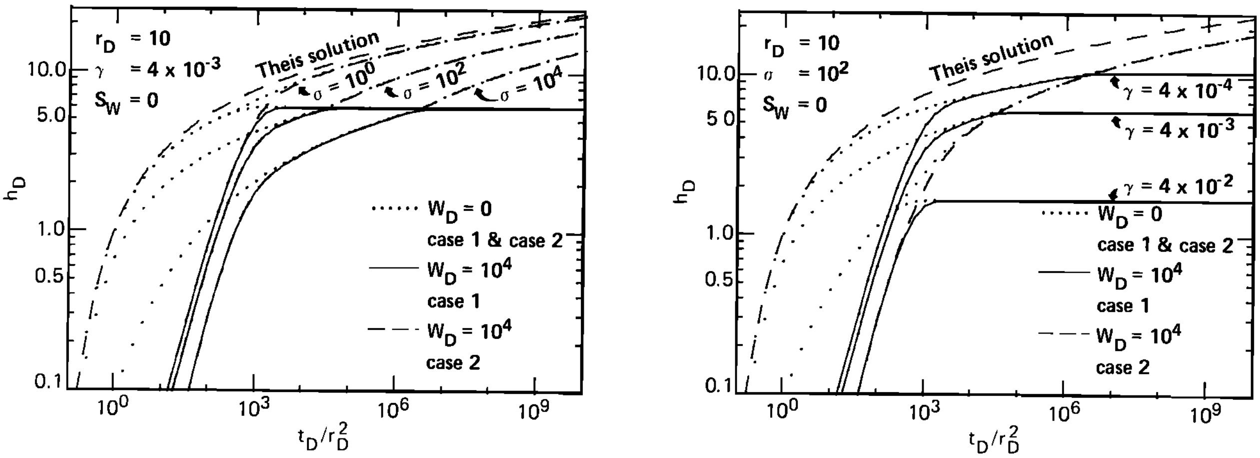

Fig. 6a

Fig. 6b

Fig. 6. Dimensionless drawdown versus dimensionless time for cases 1 and 2 in the aquifer at $r_D = 10$ for the indicated values of γ, σ, W_D, and S_w: (a) for varying values of σ holding γ fixed, (b) for varying values of γ holding σ fixed.

aquifer and in the semiconfining layers, respectively, for case 2. The same locations and parameter values as in Figures 4a and 4b are used. The early time responses, for $t_D < 10^8$, are identical in Figures 4a and 5a and in Figures 4b and 5b. Again, as z_D approaches zero, the type curves in Figure 5b approach those in Figure 5a. Hence the effects of well bore storage will become apparent in the semiconfining layers in the same way as they appear in the aquifer.

Figures 6a and 6b show the effect of varying the dimensionless hydraulic flow parameters γ and σ, for cases 1 and 2 at $r_D = 10$ assuming $S_w = 0$ and $W_D = 10^4$. Responses that would occur in the absence of well bore storage ($W_D = 0$) are also shown for comparison. In Figure 6a, γ is fixed and σ is varied, and in Figure 6b, σ is fixed and γ is varied. Examination of these figures reveals that as shown in Figures 2a and 2b, leakage derived from storage in the semiconfining layers can be completely masked by effects of well bore storage. This is especially true for small σ or large γ as shown by the early-time criterion (see equation (32)). These figures also show that the parameters W_D, γ, and σ tend to influence the character of different parts of the type curves, suggesting that unique matches with well test data may be obtained.

Figures 7a and 7b show, for case 2, how the shape of the type curves may be influenced by well bore skin. The chosen parameters are the same as were used in Figures 2a and 2b. The comparison is made for type curves assuming $S_w = 0$ and $S_w = 10$. Effects of skin are most pronounced in the pumped well (Figure 7a) where there is increased drawdown over the entire time domain. After effects of well bore storage have become negligible, the response follows the $S_w = 0$ type curve shifted vertically by a constant amount equal to $2S_w$. In an observation well (Figure 7b) the effect of skin is simply to enhance the apparent well bore storage factor by reducing drawdown at early time. It has no significant effect at large time. The effects of skin will be similar for the other two cases.

DISCUSSION

To a great extent the behavior of the type curves shown in Figures 2–7 can be anticipated by comparing early-time and late-time approximations to the Laplace transform solutions noting that the Laplace transform variable is inversely related to time. For example, effects of well bore storage can be seen to diminish with increasing time because well bore storage is

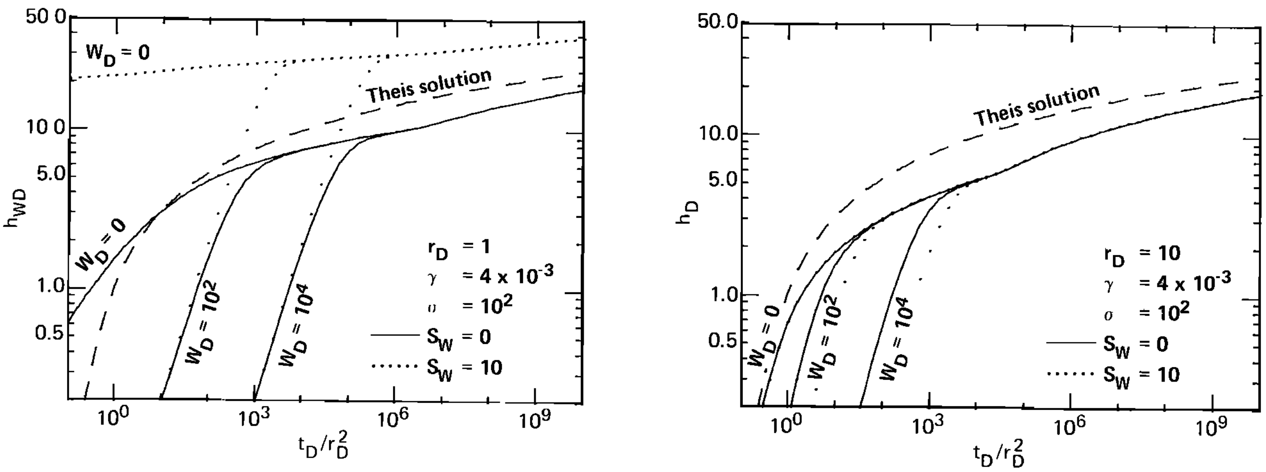

Fig. 7a

Fig. 7b

Fig. 7. Dimensionless drawdown versus dimensionless time showing the effect of skin for case 2 and for the indicated values of γ, σ, W_D, and S_w: (a) in the pumped well, (b) in the aquifer at $r_D = 10$.

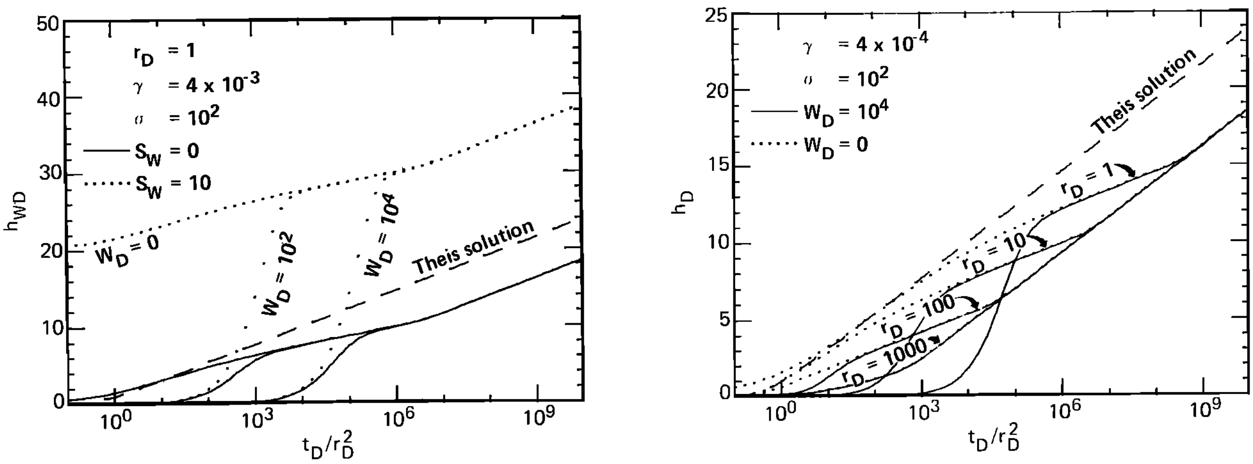

Fig. 8a

Fig. 8b

Fig. 8. Dimensionless drawdown versus dimensionless time: (*a*) semilogarithmic rendition of Figure 7*a*, (*b*) semilogarithmic rendition of Figure 5*a*.

multiplied by p and the other term in the denominator of (24) or (25) is multiplied by $(p)^{1/2}$.

The influence of the skin factor can be seen by inspection of (24). At large time, (24) reduces to

$$\bar{h}_{wD} = \frac{2K_0(x)}{pxK_1(x)} + \frac{2S_w}{p} \tag{29}$$

and for small values of the argument of the Bessel functions,

$$\bar{h}_{wD} = -\frac{\ln (p+q)}{p} + \frac{2S_w}{p} \tag{30}$$

Using the table of Laplace transforms, the inverse of (30), in the absence of leakage, is

$$h_{wD} = (\ln t_D + 2S_w + 0.5772) \tag{31}$$

This is identical, but in different notation, to an expression derived by *van Everdingen* [1953] and shows that drawdown in the production well in the absence of leakage is increased by an amount equal to $2S_w$. In the presence of leakage the effect of well bore skin is similar to that described by (31). This is demonstrated graphically in Figure 7*a* for case 2 (see also Figure 8*a*).

Early-time and late-time asymptotic approximations for drawdown in leaky aquifers were used by *Hantush* [1960, 1964] to obtain type curves for data analysis. Early-time approximations for the leakage term $\bar{q}$ in (24) or (25) derive from the fact that the hyperbolic tangents and cotangents in (26*a*), (27*a*), and (28*a*) approach unity as their arguments become large ($m > 3.16$). Late-time approximations derive from the fact that the hyperbolic tangents approach the value of their arguments and hyperbolic cotangents approach the inverse of their arguments as these arguments become small ($m < 0.316$). The range of validity for early time is thus

$$t_D < 0.1\sigma/\gamma^2 \tag{32}$$

and for late time is

$$t_D > 10\sigma/\gamma^2 \tag{33}$$

The transition from early to late time can be seen where the type curves for case 1 and case 2 depart from one another. This is clearly the point in time at which the influence of the boundaries that distinguish the two cases becomes important. By comparing this dimensionless time with computed values of t_D from (32) and (33) it can be seen that these criteria are somewhat conservative. The zone of transition from early to late time could be reduced from 2 to 1 order of magnitude without significant error. Similar conclusions were reached by *Neuman and Witherspoon* [1969*c*].

Early-time approximations for the leakage factor are the same for all three cases. From (26*a*), (27*a*), or (28*a*) they are easily shown to be

$$\bar{q} = (p)^{1/2}[\gamma'(\sigma')^{1/2} + \gamma''(\sigma'')^{1/2}] \tag{34}$$

Although termed "early" time, this approximation may, in fact, be valid for a considerable time span depending upon the parameters σ and γ. Substituting (34) into (30) and neglecting well bore skin, we obtain

$$\bar{h}_{wD} = \frac{-\ln [p + \gamma(p\sigma)^{1/2}]}{p} \tag{35}$$

Under conditions where $q(p\sigma)^{1/2}$ is large in comparison with p, a plot of drawdown versus time on semilogarithmic paper will yield a straight line with one half the slope of the Theis equation. This condition will occur when $t_D > 1/(\gamma^2\sigma)$. Hence, recalling the early-time criterion (32), the semilogarithmic half slope will occur during the time span established by the following criterion:

$$1/(\gamma^2\sigma) < t_D < 0.1\sigma/\gamma^2 \tag{36}$$

The semilogarithmic half slope is clearly evident in figures presented by *Hantush* [1960]. Its existence has also been pointed out in several papers in the recent petroleum engineering literature. A criterion similar to (36) was developed by *Serra et al.* [1983].

Late-time approximations for the leakage factor are for case 1

$$q = (\gamma')^2 + (\gamma'')^2 \tag{37}$$

for case 2

$$q = p(\sigma' + \sigma'') \tag{38}$$

and for case 3

$$q = (\gamma')^2 + p\sigma'' \tag{39}$$

By substituting (37) or (39) into (25) and allowing p to become small (large time) it can be seen, as noted by Hantush, that the arguments of the Bessel functions become independent of p and hence drawdown approaches steady state. In case 2 it can be seen, also as noted by Hantush, that drawdown must approach the Theis solution but be displaced horizontally by the factor $1 + \sigma' + \sigma''$.

Figures 8*a* and 8*b* are semilogarithmic plots of Figures 7*a* and 5*a*, respectively. They are included to illustrate graphically some of the preceding discussion. Figure 8*a* shows that after well bore storage effects become negligible, drawdown is displaced vertically by an amount equal to $2S_w$. Figure 8*b* shows that at late time the slope parallels the Theis solution and is displaced from it by a factor of $1 + 2\sigma$. In the time region defined by (36) and in the absence of well bore storage, the slope is clearly one-half that of the Theis solution.

If leakage is suspected, the reader is cautioned against use of the *Cooper and Jacob* [1946] straight line method for evaluation of aquifer properties. The line source theory of *Hantush* [1960, Figure 3] shows that drawdown plotted on semilogarithmic paper should initially follow a straight line as predicted by *Theis* [1935]. This is especially so in aquifer systems with semiconfining layers that have low hydraulic conductivity and compressibility (small $\gamma^2\sigma$). However, Figures 8*a* and 8*b* show that this portion of a semilogarithmic plot may be totally eliminated by effects of well bore storage. A straight line plot of data obtained after the effects of well bore storage have diminished may well follow a straight line, but if σ is sufficiently large so that the late-time response is not observed, the transmissivity calculated by the Cooper and Jacob method will be exaggerated by a factor of 2. Use of pumped-well data in addition to observation well data will help to establish the presence or absence of leakage because the straight line semilogarithmic plots will be displaced from one another as illustrated in Figure 8*b*. If leakage is significant and if the data all occur in the time region defined by (32), the method outlined by F. S. Riley and E. J. McClelland (U.S. Geological Survey, unpublished manuscript, 1971) is recommeded for evaluating aquifer transmissivity.

In the analysis of field data, care must be taken that a break in slope at late time such as shown in Figure 8 not be misinterpreted as being due to the presence of an impermeable barrier. The presence of an impermeable barrier would also result in a change in slope by a factor of 2. More than one observation well may be needed to distinguish the two situations. If the cause is an impermeable boundary, the break in slope will occur at a time that is dependent upon observation well location. The location of the barrier may possibly be identified by using the methods outlined by *Ferris et al.* [1962, pp. 161–166]. If the cause is due to case 2 described here, the break in slope, for a homogeneous and isotropic aquifer, should occur at the same time at all locations. This conclusion can be inferred from Figure 8*b*.

In order to obtain as much information as possible about a given leaky aquifer system it is highly desirable to obtain not only observation well data but also pumped-well data and semiconfining layer piezometer data. Drawdown in the pumped well, even though irregular because of discharge fluctuations, should be helpful in determining the presence or absence of leakage when compared with observation well data.

It was pointed out in the previous section that early-time response, including effects of leakage derived from storage in semiconfining layers, can be obliterated by effects of well bore storage in a large-diameter well. Therefore care should be taken in interpreting the results of a type curve match to the standard Theis solution. Such a match could yield an erroneously large value of aquifer storativity. Apart from changing the standpipe diameter, one way to reduce or effectively eliminate the masking effect of well bore storage is to isolate the aquifer of interest with hydraulic packers and repeat the pump test under pressurized conditions. Because well bore storage C will then be due to fluid compressibility rather than changing water levels in the well (see the discussion following (4)), the dimensionless well bore storage parameter may be reduced by 4 to 5 orders of magnitude. This approach was proposed by *Warren and Root* [1963] to circumvent the masking effects of "afterflow" or well bore storage during pressure buildup. If effects of leakage are apparent in the well test data, it may be possible, through a systematic process of trial and error, to obtain a data match that is unique to the problem at hand. This is because the dimensionless parameters W_D, σ, and γ tend to influence different parts of the type curves.

It should be clear from the type curves shown in Figures 2–7 that long-term pump tests are necessary if it is to be determined which of the three cases, if any, applies to a given field situation. This is true even if semiconfining layer piezometer data are available. Long-term data should be used with caution, however, because of the likeihood of interference from extraneous effects and boundary conditions. Short-term tests in leaky aquifers have the drawback that many of the early-time type curves appear similar to one another and therefore cannot be used as a stand-alone diagnostic tool. The practicing groundwater hydrologist should be wary of these limitations and seek out corroborating geologic and hydrologic evidence.

Should type curves, such as shown in Figure 2–7, be needed for data analysis, they can be readily generated using the *Stehfest* [1970] algorithm and a plotting device. The Fortran IV computer code used for this paper is available from the author upon request.

APPENDIX

The coupled boundary value problems defined by (1)–(14) can be put into nondimensional form by use of (15)–(23).

For the aquifer in which the lower semiconfining layer is neglected, the nondimensional governing differential equation is

$$\frac{\partial^2 h_D}{\partial r_D{}^2} + \frac{1}{r_D}\frac{\partial h_D}{\partial r_D} = \frac{\partial h_D}{\partial t_D} + w_D' \qquad r_D \geq 1 \tag{A1}$$

where

$$w_D' = -\frac{K' r_w{}^2}{Kbb'}\left(\frac{\partial h_D'}{\partial z_D'}\right)_{z_{D'}=0} = -\gamma'^2\left(\frac{\partial h_D'}{\partial z_D'}\right)_{z_{D'}=0}$$

The initial condition is

$$h_D = 0 \qquad r_D > 1 \tag{A2}$$

The boundary conditions are

$$h_D = 0 \qquad r_D = \infty \tag{A3}$$

$$W_D \frac{\partial h_{wD}}{\partial t_D} - \frac{\partial h_D}{\partial r_D} = 2 \qquad r_D = 1 \tag{A4}$$

where

$$h_{wD} = h_D - S_w \frac{\partial h_D}{\partial r_D}$$

For the overlying, semiconfining layer the nondimensional governing differential equation is

$$\frac{\partial^2 h_D'}{\partial z_D'^2} = \frac{\sigma'}{(\gamma')^2} \frac{\partial h_D'}{\partial t_D} \qquad 0 \le z_D' \le 1 \tag{A5}$$

The initial condition is

$$h_D' = 0 \qquad 0 \le z_D' \le 1 \tag{A6}$$

the boundary conditions are

$$h_D' = h_D \qquad z_D' = 0 \tag{A7}$$

and, for case 1,

$$h_D' = 0 \qquad z_D' = 1 \tag{A8}$$

or, for case 2,

$$\frac{\partial h_D'}{\partial z_D'} = 0 \qquad z_D' = 1 \tag{A9}$$

By performing the Laplace transform procedure on (A1)–(A9) the subsidiary boundary value problems are obtained.

The subsidiary differential equation for the aquifer is

$$\frac{\partial^2 \bar{h}_D}{\partial r_D^2} + \frac{1}{r_D} \frac{\partial \bar{h}_D}{\partial r_D} = p\bar{h}_D + \bar{w}_D' \qquad r_D \ge 1 \tag{A10}$$

where

$$\bar{w}_D' = -\gamma'^2 \left(\frac{\partial \bar{h}_D'}{\partial z_D'}\right)_{z_D'=0}$$

and the subsidiary boundary conditions are

$$\bar{h}_D = 0 \qquad r_D = \infty \tag{A11}$$

$$W_D \bar{h}_{wD} - \frac{1}{p} \frac{\partial \bar{h}_D}{\partial r_D} = \frac{2}{p^2} \qquad r_D = 1 \tag{A12}$$

where

$$\bar{h}_{wD} = \bar{h}_D - S_w \frac{\partial \bar{h}_D}{\partial r_D} \tag{A13}$$

The subsidiary differential equation for the overlying semiconfining layer is

$$\frac{\partial^2 \bar{h}_D'}{(\partial z_D')^2} = \frac{\sigma' p}{(\gamma')^2} \bar{h}_D' = (m')^2 \bar{h}_D' \qquad 0 \le z_D' \le 1 \tag{A14}$$

and the subsidiary boundary conditions are

$$\bar{h}_D' = \bar{h}_D \qquad z_D' = 0 \tag{A15}$$

and, for case 1,

$$\bar{h}_D' = 0 \qquad z_D' = 1 \tag{A16}$$

or, for case 2,

$$\frac{\partial \bar{h}_D'}{\partial z_D'} = 0 \qquad z_D' = 1 \tag{A17}$$

The general solution to (A14) is

$$\bar{h}_D' = A \cosh(z_D' m') + B \sinh(z_D' m') \tag{A18}$$

The particular solutions, obtained by applying (A15) and (A16) or (A17), are, for case 1,

$$\bar{h}_D' = \bar{h}_D \sinh[(1 - z_D')m']/\sinh(m') \tag{A19}$$

or, for case 2,

$$\bar{h}_D' = \bar{h}_D \cosh[(1 - z_D')m']/\cosh(m') \tag{A20}$$

Substitution of (A19) or (A20) into (A10) yields

$$\frac{\partial^2 \bar{h}_D}{\partial r_D^2} + \frac{1}{r_D} \frac{\partial \bar{h}_D}{\partial r_D} = \bar{h}_D(p + \bar{q}) \tag{A21}$$

where for case 1,

$$\bar{q} = (\gamma')^2 m' \coth(m') \tag{A22}$$

and for case 2,

$$\bar{q} = (\gamma')^2 m' \tanh(m') \tag{A23}$$

The same development can be applied to a lower semiconfining layer leading to (26)–(28) for the three cases.

The general solution of (A21) is

$$\bar{h}_D = A I_0(r_D x) + B K_0(r_D x) \tag{A24}$$

where $x = (p + \bar{q})^{1/2}$. Since by (A11), $\bar{h}_D$ is bounded, $A = 0$ and

$$\bar{h}_D = B K_0(r_D x) \tag{A25}$$

Applying (A12),

$$B = \frac{2/p - pW_D \bar{h}_{wD}}{x K_1(x)} \tag{A26}$$

Applying (A13) and by algebraic manipulation the particular solutions are

$$\bar{h}_{wD} = \frac{2[K_0(x) + xS_w K_1(x)]}{p\{pW_D[K_0(x) + xS_w K_1(x)] + xK_1(x)\}} \tag{A27}$$

$$\bar{h}_D = \frac{2K_0(r_D x)}{p\{pW_D[K_0(x) + xS_w K_1(x)] + xK_1(x)\}} \tag{A28}$$

Equations (A27) and (A28) are the solutions (24) and (25).

NOTATION

- b thickness of a given layer, L.
- C well bore storage, L^2.
- h hydraulic head in a given layer, L.
- h_i initial hydraulic head in a given layer, L.
- h_D dimensionless drawdown in a given layer.
- h_w hydraulic head in a pumped well, L.
- h_{wD} dimensionless drawdown in pumped well.
- I_0 modified Bessel function of the first kind and order zero.
- K_0 modified Bessel function of the second kind and order zero.
- K_1 modified Bessel function of the second kind and order unity.
- K hydraulic conductivity of a given layer, LT^{-1}.
- p Laplace transform variable.
- Q well discharge rate, L^3T^{-1}.
- $\bar{q}$ relating function for leakage from a given semiconfining layer.
- r radial coordinate originating at the center of pumped well, L.
- r_D dimensionless radial distance.
- r_w effective radius of pumped well, L.
- S_s specific storage of a given layer, L^{-1}.
- S_w dimensionless coefficient of well bore skin.
- t time since start of pumping, T.

t_D dimensionless time.
w source function for leakage from a given semiconfining layer, L^{-1}.
W_D dimensionless well bore storage coefficient.
z vertical coordinate originating at base of lower semiconfining layer, L.
z_D dimensionless vertical distance.

Single primes denote the upper semiconfining layer; double primes denote the lower semiconfining layer.

Dimensionless groupings for reservoir parameters

$$m' = \frac{(\sigma' p)^{1/2}}{\gamma'} \qquad m'' = \frac{(\sigma'' p)^{1/2}}{\gamma''}$$

$$\gamma' = r_w\left(\frac{K'}{Kbb'}\right)^{1/2} \qquad \gamma'' = r_w\left(\frac{K''}{Kbb''}\right)^{1/2}$$

$$\sigma' = \frac{S_s'b'}{S_s b} \qquad \sigma'' = \frac{S_s''b''}{S_s b}$$

$$\beta' = \frac{\gamma'}{4}(\sigma')^{1/2} r_D \qquad \beta'' = \frac{\gamma''}{4}(\sigma'')^{1/2} r_D$$

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A. F. Moench, Water Resources Division, U.S. Geological Survey, 345 Middlefield Road, Menlo Park, CA 94025.

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